

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

PILLAR BIOSCIENCES, INC.,
Petitioner,

v.

SWIFT BIOSCIENCES, INC.,
Patent Owner.

IPR2021-00401
Patent 10,316,359 B2

Before SHERIDAN K. SNEDDEN, ZHENYU YANG,
and CHRISTOPHER M. KAISER, *Administrative Patent Judges*.

KAISER, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

INTRODUCTION

A. Background

Pillar Biosciences, Inc. (“Petitioner”) filed a Petition (Paper 2, “Pet.”) requesting an *inter partes* review of claims 1–6 of U.S. Patent No. 10,316,359 B2 (Ex. 1001, “the ’359 patent”). Swift Biosciences, Inc. (“Patent Owner”) filed a Preliminary Response. Paper 6 (“Prelim. Resp.”).

On July 19, 2021, we instituted *inter partes* review of claims 1–6 of the ’359 patent on all grounds set forth in the Petition. Paper 7 (“Inst. Dec.”). After institution of trial, Patent Owner filed a Corrected Response (Paper 16, “PO Resp.”), Petitioner filed a Reply (Paper 20), and Patent Owner filed a Sur-Reply (Paper 22). In addition, Patent Owner filed a Motion to Exclude Evidence (Paper 26), Petitioner filed an Opposition (Paper 27), and Patent Owner filed a Reply (Paper 28). At the request of both parties, we held an oral hearing, the transcript of which has been entered into the record. Paper 32 (“Tr.”).

We have jurisdiction under 35 U.S.C. § 6. This is a Final Written Decision under 35 U.S.C. § 318(a) as to the patentability of the challenged claim of the ’359 patent. For the reasons discussed below, we determine Petitioner has established by a preponderance of the evidence that each of claims 1–6 of the ’359 patent is unpatentable.

B. Related Matters

The parties identify no matters related to the present proceeding. Pet. 4; Paper 4, 1.

C. The Instituted Grounds of Unpatentability

Petitioner contends that claims 1–6 of the ’359 patent are unpatentable based on the following grounds (Pet. 28–67):¹

Claims Challenged	35 U.S.C. § ²	References/Basis
1–6	102	Xie ³
1–6	102	Lao ⁴
1–6	103	Gardner, ⁵ Lao

D. The ’359 Patent

The ’359 patent, titled “Methods for Multiplex PCR,” issued on June 11, 2019. Ex. 1001, codes (45), (54). The patent relates to “[m]ethods for the preparation of PCR reaction mixtures and for performing multiplex PCR amplification which limit the production of non-target amplicons.” *Id.* at code (57).

Figure 44A of the ’359 patent, reproduced below, illustrates the features of the claimed method.

¹ Petitioner also relies on a declaration from Yun Bao, Ph.D. Ex. 1002.

² The earliest priority date claimed by the ’359 patent is January 31, 2014, and Petitioner argues that the patent is entitled to priority only to November 11, 2014. Pet. 20–22; Ex. 1001, codes (22), (60), (63). In either case, the application leading to the ’359 patent has an effective filing date after March 16, 2013, so patentability is governed by the versions of 35 U.S.C. §§ 102 and 103 as amended by the Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112–29, 125 Stat. 284 (2011).

³ US 2014/0200146 A1, published July 17, 2014 (Ex. 1018).

⁴ US 2009/0291475 A1, published Nov. 26, 2009 (Ex. 1014).

⁵ Shea N. Gardner, Crystal J. Jaing, Maher M. Elsheikh, José Peña, David A. Hysom, & Monica K. Borucki, *Multiplex Degenerate Primer Design for Targeted Whole Genome Amplification of Many Viral Genomes*, ADVANCES IN BIOINFORMATICS, vol. 2014, at 1–8 (Ex. 1016).

FIG. 44A

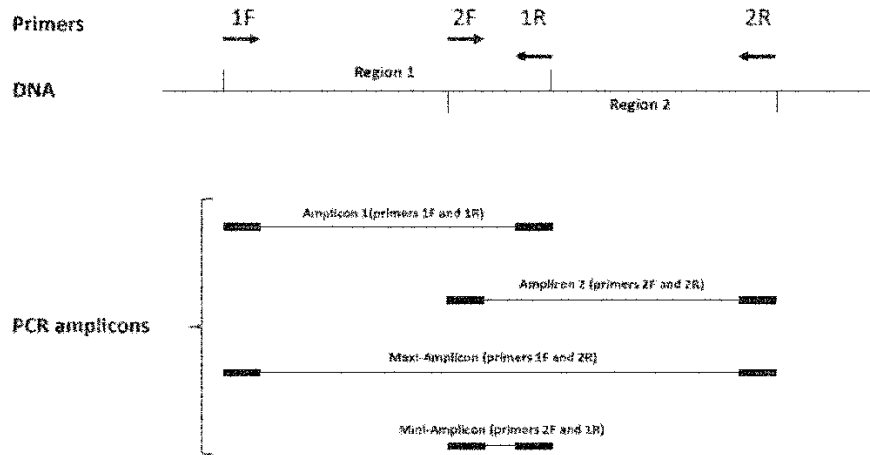


Figure 44A depicts two sets of primer pairs, each with a forward and a reverse primer. *Id.* at Fig. 44A. One set of primer pairs, set 1, contains forward primer 1F and reverse primer 1R, chosen to target the DNA sequence at Region 1. *Id.* The other set of primer pairs, set 2, contains forward primer 2F and reverse primer 2R, chosen to target the DNA sequence at Region 2. *Id.* Because the target regions overlap, the primers can produce four PCR amplicons, each of which is depicted in Figure 44A. First, primers 1F and 1R can produce Amplicon 1. *Id.* Second, primers 2F and 2R can produce Amplicon 2. *Id.* Third, primers 1F and 2R (“the two distal primers”) can produce a “Maxi-Amplicon.” *Id.* at 8:51–55, Fig. 44A. Finally, primers 2F and 1R (“the two proximal primers”) can produce a “Mini-Amplicon.” *Id.* at 8:51–57, Fig. 44A.

The “mini amplicon” produced by primers 2F and 1R can “dominate amplification reactions due to [its] short length and ease of amplification,” and the ’359 patent seeks to avoid this effect. *Id.* at 8:57–60. According to

the patent, “most methods separate overlapping primer pairs into two tubes” in order to “avoid having the mini amplicon dominate the multiplexed PCR reaction,” but this “doubles the workload and required DNA input quantity.” *Id.* at 8:57–63. The method of the ’359 patent “enables overlapping amplicons to be created in a single tube.” *Id.* at 8:63–64.

To accomplish this, the ’359 patent describes using “pairs of target-specific primers” with “a universal truncated [next-generation sequencing] adapter sequence at their 5’ termini.” *Id.* at 6:47–50, Figs. 40, 41. “[D]ue to the presence of the universal sequence at each terminus, the short mini amplicon will be subject to stable secondary structure which results in less efficient amplification by the single universal primer.” *Id.* at 8:65–9:1. “Therefore, even if the mini amplicon is produced during the initial target specific PCR cycles, it will not be efficiently amplified.” *Id.* at 9:1–3.

E. Illustrative Claim

Claims 1–6 of the ’359 patent are challenged. Claims 1 and 3 are independent, and claim 1 is illustrative; it recites:

1. A method of multiplex PCR amplification of a target nucleic acid substrate comprising the steps of:
 - (i) combining a plurality of target-specific primers with the target nucleic acid substrate to yield a single polymerase chain reaction (PCR) reaction mixture, wherein the plurality of target-specific primers comprise a first forward primer, a second forward primer, a first reverse primer and a second reverse primer, wherein each of the first and second forward and reverse primers comprise a 3’ complementary sequence that is complementary to the target nucleic acid substrate and a 5’ noncomplementary sequence that is not complementary to the target nucleic acid substrate,

wherein the 3' complementary sequence for each of the first and second forward and reverse primers is different;

(ii) subjecting the PCR reaction mixture to a multiplex polymerase chain reaction thereby generating at least three amplicons, wherein the at least three amplicons comprise a first amplicon produced by the first forward primer and the first reverse primer, a second amplicon produced by the second forward primer and the second reverse primer, and a third amplicon produced by the second forward primer and the first reverse primer, wherein at least a portion of the 5' noncomplementary sequence of the second forward primer and the first reverse primer is the same such that each strand of the third amplicon comprises a 3' end and a 5' end that are complementary to each other, wherein the third amplicon possesses overlapping sequence with the first and second amplicons, wherein the first amplicon possesses overlapping sequence with the second amplicon, wherein when the third amplicon is denatured, each strand of the third amplicon forms a secondary structure as a result of the 3' end being complementary to the 5' end, and wherein the secondary structure is stable during a primer annealing step of the multiplex polymerase chain reaction.

Ex. 1001, 133:56–135:12.

ANALYSIS

A. Claim Construction

In an *inter partes* review, we construe a claim in an unexpired patent “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” 37 C.F.R. § 42.100(b) (2020). “[T]he ordinary and customary meaning of a claim term is the meaning that the term would have to a person of ordinary skill in the art in question at the time of the

invention.” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1313 (Fed. Cir. 2005) (en banc). “Importantly, the person of ordinary skill in the art is deemed to read the claim term not only in the context of the particular claim in which the disputed term appears, but in the context of the entire patent, including the specification.” *Id.* “[W]e need only construe terms ‘that are in controversy, and only to the extent necessary to resolve the controversy.’” *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

Petitioner proposes constructions for five terms: “third amplicon,” “target-specific primer,” “complementary sequence,” “noncomplementary sequence,” and “PCR.” Pet. 22–28. Patent Owner disputes only the proposed constructions of “third amplicon” and “target-specific primer.” PO Resp. 9–17. As discussed in more detail below, the parties do not dispute that the combination of Gardner and Lao teaches or suggests the subject matter of the challenged claims, so there is no need for us to resolve the parties’ claim-construction dispute as to either of the terms on which they disagree in order to resolve the question of obviousness. In addition, although the parties’ dispute over the construction of “target-specific primer” is germane to whether Petitioner has proven the anticipation of the challenged claims by either Lao or Xie, we do not reach the anticipation grounds. Thus, we need not, and we do not, construe any claim terms in order to resolve this proceeding.

B. Asserted Obviousness over Gardner and Lao

Petitioner argues that claims 1–6 would have been obvious over the combination of Gardner and Lao. Pet. 54–67.

1. *Gardner*

Gardner relates to “designing multiplex sets of degenerate sequencing primers to tile overlapping amplicons across multiple whole genomes.” Ex. 1016, 1. The process of Gardner has the following steps: “split a multiple sequence alignment into overlapping regions, and for each region design a degenerate multiplex set of primers that in combination amplify that region in all strains with as few primers as possible.” Ex. 1016, 3. This process “can be run in separate singleplex reactions for each split region, or, alternatively, primers for all regions can be combined in a large multiplex after the large set is checked for primer dimers that could occur between primers from different regions.” *Id.* Gardner teaches that “[c]ombining primers for all regions in multiplex should facilitate whole genome amplification in a single reaction” and “may yield longer amplicons from the reaction of forward and reverse primers from different parts.” *Id.* at 3–4. Because “primers of overlapping regions can also produce amplicons shorter . . . than the desired amplicon . . . , a step to remove short amplicons before sequencing may be desired.” *Id.* at 4. In Gardner’s “experimental test” of its method, “the primers from parts 0, 2, and 4 were combined in one reaction and the primers from parts 1 and 3 were combined in another, so that short products would not be produced.” *Id.* Gardner reports the results of a test of its method on the “murine hepatitis virus” in which “[e]ach region [was] amplified using 3 forward primers and 3 reverse primers.” *Id.* at Fig. 2; *see* Ex. 1019, Table S1.

2. *Lao*

Lao “relates to methods and compositions for amplifying nucleic acid sequences.” Ex. 1014 ¶ 3. In part, Lao discloses “an event or state in which

a portion of a nucleic acid strand is hybridized to another portion of itself,” which Lao refers to as “self-hybridizing.” *Id.* ¶ 69. This “self-hybridized structure [may only be] formed for the shorter insert sections,” allowing for “selective amplification of longer insert sections over shorter insert sections.” *Id.* ¶ 116. This happens because the shorter insert sections “take themselves out of a reaction” through self-hybridization. *Id.* Lao’s process can be used to amplify target portions of DNA, including amplifying “more than one locus . . . in one reaction.” *Id.* ¶ 166.

3. *Analysis*

Petitioner argues that the combination of Gardner and Lao teaches or suggests the subject matter of claims 1–6 and that a person of ordinary skill in the art would have had a reason to combine the teachings of these references with a reasonable expectation of success. Pet. 54–67; Reply 2–20. Patent Owner does not dispute that each individual limitation may be found in the prior art but disagrees that Petitioner has sufficiently shown a reason to combine the references so as to arrive at the claimed invention with a reasonable expectation of success. PO Resp. 26–46; Sur-Reply 2–22.

a. Claim 1

As stated above, claim 1 recites “[a] method of multiplex PCR amplification of a target nucleic acid substrate.” Ex. 1001, 133:56–57. Petitioner argues that Gardner teaches this portion of claim 1. Pet. 57. Patent Owner does not dispute Petitioner’s argument that the prior art teaches this, or any other, portion of claim 1. PO Resp. 26–46; Sur-Reply 2–22. Gardner teaches “multiplex PCR,” with examples in which multiple loci within a DNA substrate are amplified. Ex. 1016, 4, 7. Accordingly,

Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this portion of claim 1.

Claim 1 also recites “combining a plurality of target-specific primers with the target nucleic acid substrate to yield a single polymerase chain reaction (PCR) reaction mixture.” Ex. 1001, 133:58–60. Petitioner argues that Gardner teaches this limitation. Pet. 57–58. Gardner teaches using forward and reverse primers for each locus to be amplified, amplifying multiple loci in a single reaction, and combining the primers with the target nucleic acid. Ex. 1016, 4, 7, Fig. 2. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

Claim 1 recites “wherein the plurality of target-specific primers comprise a first forward primer, a second forward primer, a first reverse primer and a second reverse primer.” Ex. 1001, 133:60–63. Petitioner argues that Gardner teaches this limitation. Pet. 59. Gardner teaches using multiple sets of primers, at least one set for each locus to be amplified, with each set comprising a forward primer and a reverse primer. Ex. 1016, Fig. 2; Ex. 1019, Table S1. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

Claim 1 recites “wherein each of the first and second forward and reverse primers comprise a 3′ complementary sequence that is complementary to the target nucleic acid substrate and a 5′ noncomplementary sequence that is not complementary to the target nucleic acid substrate.” Ex. 1001, 133:63–134:55. Petitioner argues that the combination of Gardner and Lao teaches or suggests this limitation.

Pet. 59–60. Specifically, Petitioner argues that Gardner teaches the “3′ complementary sequence that is complementary to the target nucleic acid substrate” and Lao teaches the “5′ noncomplementary sequence that is not complementary to the target nucleic acid substrate.” *Id.* Gardner’s primers contain 3′ sequences that are between 88 and 100 percent complementary to their target regions. Ex. 1019, Table S1. Lao’s primers include a noncomplementary region at the 5′ region of the primer that is noncomplementary with respect to the target nucleic acid sequence. Ex. 1014 ¶¶ 44, 47, 52, 54, 81. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

Claim 1 also recites “wherein the 3′ complementary sequence for each of the first and second forward and reverse primers is different.” Ex. 1001, 134:55–57. Petitioner argues that Gardner teaches this limitation. Pet. 60–61. Each of Gardner’s primers is designed with a different sequence to amplify different loci. Ex. 1016, 7; Ex. 1019, Table S1. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

Claim 1 recites “subjecting the PCR reaction mixture to a multiplex polymerase chain reaction thereby generating at least three amplicons.” Ex. 1001, 134:58–60. Petitioner argues that Gardner teaches this limitation. Pet. 61. Gardner teaches generating amplicons for each target locus; in one of Gardner’s reaction tubes, amplicons are generated for loci B, D, F, and H, meaning that at least four amplicons are generated. Ex. 1016, 7. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

Claim 1 also recites

wherein the at least three amplicons comprise a first amplicon produced by the first forward primer and the first reverse primer, a second amplicon produced by the second forward primer and the second reverse primer, and a third amplicon produced by the second forward primer and the first reverse primer.

Ex. 1001, 134:60–65. Petitioner argues that the combination of Gardner and Lao teaches or suggests this limitation. Pet. 61–62. Gardner teaches using multiple sets of forward and reverse primers, some of which overlap.

Ex. 1016, Fig. 2. Gardner also teaches that, when “primers of overlapping regions” are used in the same reaction tube, this can “produce amplicons shorter . . . than the desired amplicon,” and Gardner suggests splitting the multiplex PCR reactions into separate tubes to avoid this problem. *Id.* at 4. But Gardner also teaches that there are benefits to running all the PCR reactions simultaneously in the same tube, and Lao provides a means to do so, permitting the short amplicons to be generated without the problem that Gardner describes. *Id.*; Ex. 1014 ¶¶ 116, 166. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

Claim 1 recites “wherein at least a portion of the 5’ noncomplementary sequence of the second forward primer and the first reverse primer is the same such that each strand of the third amplicon comprises a 3’ end and a 5’ end that are complementary to each other.” Ex. 1001, 134:66–135:3. Petitioner argues that Lao teaches or suggests this limitation. Pet. 62–63. Lao teaches that short amplicons formed with its process can form a “self-hybridized structure.” Ex. 1014 ¶ 116. This

structure forms when “a portion of a nucleic acid strand is hybridized to another portion of itself,” such as when there is “effective hybridization of the linear primer to at least a portion of the linear primer complement.” *Id.* ¶ 69, Fig. 5. This occurs because Lao’s amplicons have complementary 5’ and 3’ sequences at opposite ends of each strand. Ex. 1002 ¶ 203.

Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

Claim 1 also recites “wherein the third amplicon possesses overlapping sequence with the first and second amplicons, wherein the first amplicon possesses overlapping sequence with the second amplicon.” Ex. 1001, 135:3–6. Petitioner argues that the combination of Gardner and Lao teaches this limitation. Pet. 63–64. When Gardner’s reactions are all performed in the same tube, they produce sets of two intended amplicons that overlap one another, as well as an additional short amplicon that overlaps both of the intended amplicons. Ex. 1016, 4, Fig. 2. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

Claim 1 recites “wherein when the third amplicon is denatured, each strand of the third amplicon forms a secondary structure as a result of the 3’ end being complementary to the 5’ end.” Ex. 1001, 135:7–10. Petitioner argues that Lao teaches this limitation. Pet. 64–65. Lao teaches that shorter amplicons can form a loop structure by self-hybridizing. Ex. 1014 ¶ 116, Fig. 5. Lao also teaches that this self-hybridization occurs as a result of the 3’ end being complementary to the 5’ end. *Id.* ¶ 69. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

Claim 1 also recites “wherein the secondary structure is stable during a primer annealing step of the multiplex polymerase chain reaction.”

Ex. 1001, 135:10–12. Petitioner argues that Lao teaches this limitation.

Pet. 65. Lao teaches that shorter amplicons “take themselves out of a reaction” because they self-hybridize to form loops more quickly than longer amplicons. Ex. 1014 ¶ 116. Lao also teaches that the loop structure formed by the short amplicons “is too stable to permit efficient hybridization and extension during PCR.” *Id.* ¶ 138. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests this limitation.

b. Claim 3

Claim 3 is quite similar to claim 1. *Compare* Ex. 1001, 133:56–135:12, *with* Ex. 1001, 135:18–136:22. The only differences between the claims are that (1) claim 3 recites “[a] method for preparing a PCR reaction mixture for multiplex PCR amplification” rather than “[a] method of multiplex PCR amplification”; and (2) claim 3 recites “the target specific primers are designed in a manner capable of producing at least three amplicons” rather than “subjecting the PCR reaction mixture to a multiplex polymerase chain reaction thereby generating at least three amplicons.” *Id.*

Petitioner relies on its arguments with respect to claim 1 to show that the combination of Gardner and Lao teaches or suggests the subject matter of claim 3. Pet. 40–41. Patent Owner does not dispute Petitioner’s argument that the prior art teaches the subject matter of claim 3. PO Resp. 26–46; Sur-Reply 2–22. Thus, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches the subject matter of claim 3.

c. Claims 2 and 4

Claims 2 and 4 depend from claims 1 and 3, respectively, and add a limitation requiring that “the at least three amplicons further comprise a fourth amplicon produced by the first forward primer and the second reverse primer, and wherein the fourth amplicon overlaps the first, second, and third amplicon.” Ex. 1001, 135:13–17, 136:23–27. Petitioner argues that the combination of Gardner and Lao teaches or suggests this limitation. Pet. 65–67. Patent Owner does not dispute this argument. PO Resp. 26–46; Sur-Reply 2–22. When Gardner’s reactions are all performed in the same tube, they produce sets of two intended amplicons that overlap one another, as well as an additional short amplicon that overlaps both of the intended amplicons. Ex. 1016, 4, Fig. 2. Gardner also states that performing multiplex PCR reactions in a single tube “may yield longer amplicons from the reaction of forward and reverse primers from different parts.” *Id.* at 4. Moreover, Dr. Bao testifies that, “[w]hen all of Gardner’s primers are combined in the same reaction, the claimed fourth amplicon will be produced.” Ex. 1002 ¶¶ 209–210. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches the subject matter of claims 2 and 4.

d. Claims 5 and 6

Claims 5 and 6 depend from claims 1 and 3, respectively, and add a limitation requiring that “the entire 5’ non-complementary sequence of the second forward primer and the first reverse primer is the same.” Ex. 1001, 136:28–33. Petitioner argues that Lao teaches this limitation. Pet. 67. Patent Owner does not dispute this argument. PO Resp. 26–46; Sur-Reply 2–22. Lao teaches that the 5’ noncomplementary sequence is the same for

all of its primers. Ex. 1014 ¶ 53. Accordingly, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches the subject matter of claims 5 and 6.

e. Reason to Combine and Reasonable Expectation of Success

As discussed above, Petitioner has shown by the preponderance of the evidence that the combination of Gardner and Lao teaches or suggests every limitation of the challenged claims. “An invention is not obvious just ‘because all of the elements that comprise the invention were known in the prior art.’” *Broadcom Corp. v. Emulex Corp.*, 732 F.3d 1325, 1335 (Fed. Cir. 2013) (quoting *Power-One, Inc. v. Artesyn Techs., Inc.*, 599 F.3d 1343, 1351 (Fed. Cir. 2010)). Instead, “a finding of obviousness at the time of invention requires a ‘plausible rational[e] as to why the prior art references would have worked together.’” *Id.* (quoting *Power-One*, 599 F.3d at 1352). This rationale should include “a motivation to combine accompanied by a reasonable expectation of achieving what is claimed in the patent-at-issue.” *Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367 (Fed. Cir. 2016).

(1) Reason to Combine

Petitioner argues that a person of ordinary skill in the art would have been motivated to combine the teachings of Gardner with those of Lao (1) because Gardner suggests such a combination expressly, and (2) in order to achieve Gardner’s benefit of avoiding overlapping amplicons while also achieving Lao’s benefit of saving time and cost due to avoiding having to run separate reactions. Pet. 54–56; *see* Reply 3–12. Patent Owner disagrees with both of these arguments. PO Resp. 26–37; Sur-Reply 4–14.

We first consider Petitioner’s argument for an express reason to combine the teachings of Gardner with those of Lao. Gardner discusses how to carry out reactions for each of its several primers. Ex. 1016, 3–4. Specifically, Gardner teaches that “[t]he primers can be run in separate singleplex reactions for each split region, or, alternatively, primers for all regions can be combined in a large multiplex after the large set is checked for primer dimers that could occur between primers from different regions.” *Id.* at 3. With respect to the multiplex option, Gardner teaches that “[c]ombining primers for all regions in multiplex should facilitate whole genome amplification in a single reaction.” *Id.* at 3–4; *see also id.* at 2 (“The method here . . . seeks to automatically design a minimal, degenerate set of multiplex compatible primers to amplify all the strains for a given region in a single reaction.”). According to Gardner, this approach provides several benefits, including that “[i]t may yield longer amplicons . . . and should facilitate assembly across amplified regions.” *Id.* at 4. These benefits come at a cost: “since primers of overlapping regions can also produce amplicons shorter . . . than the desired amplicon length . . . , a step to remove short amplicons before sequencing may be desired.” *Id.* Gardner then suggests avoiding this problem by carrying out two separate multiplex reactions, with “the primers from parts 0, 2, and 4 . . . combined in one reaction and the primers from parts 1 and 3 . . . combined in another, so that short products would not be produced.” *Id.*

Petitioner argues that this disclosure of Gardner teaches (1) that carrying out a single multiplex reaction is desirable and (2) that such a multiplex reaction could be accomplished if some way could be found to remove the short amplicons that would be produced. Pet. 54–55. According

to Petitioner, these two teachings of Gardner constitute an express suggestion to combine Gardner's single-reaction method with a method of removing short amplicons, such as the method taught by Lao. *Id.*; Reply 5–7.

Patent Owner argues that Gardner teaches its own solution for avoiding the production of small amplicons. PO Resp. 28–30. According to Patent Owner, by splitting the single multiplex reaction into two separate reactions, and thereby avoiding the production of any small amplicons, Ex. 1016, 4, Gardner solves any short-amplicon problem, leaving no need for a combination with Lao's method of removing short amplicons. PO Resp. 28–30. Further, Patent Owner argues that “Gardner is quite opposed to a single reaction and even warns about additional complexity and steps.” Sur-Reply 6–7.

We agree with Petitioner. Although Gardner does, as Patent Owner notes, teach a method of avoiding the production of short amplicons, it does so only by carrying out two separate reactions. Ex. 1016, 4 (suggesting combining “the primers from parts 0, 2, and 4 . . . in one reaction and the primers from parts 1 and 3 . . . in another, so that short products would not be produced”). But Gardner makes clear that carrying out a single multiplex reaction would be preferable. *Id.* at 2 (“The method here . . . seeks to automatically design a minimal, degenerate set of multiplex compatible primers to amplify all the strains for a given region in a single reaction.”), 4 (carrying out “whole genome amplification in a single reaction . . . may yield longer amplicons . . . and should facilitate assembly across amplified regions”). Thus, Gardner favors finding a way to carry out a single reaction rather than settling for its collection of separate reactions.

Gardner teaches that, without “staggering the primer sets into different multiplex reactions,” the short amplicons produced by “overlapping primer sets would dominate the PCR reaction.” *Id.* at 6. Patent Owner argues that this shows that Gardner disapproves of any means of controlling short amplicons other than its multiple-reaction approach and is “quite opposed to a single reaction.” Sur-Reply 6–7. We disagree. As noted above, Gardner also teaches that “a step to remove short amplicons before sequencing may be desired.” Ex. 1016, 4. Gardner’s multi-reaction solution avoids production of short amplicons in the first place, and amplicons that are never produced do not need to be removed. Thus, the fact that Gardner teaches the removal of short amplicons means that Gardner does not disapprove of the use of solutions other than its multi-reaction approach.

Instead, Gardner considers its multi-reaction approach to be a compromise between the goal of avoiding problems caused by the production of short amplicons and the goal of minimizing the number of reactions carried out. Gardner’s teaching that adding “a step to remove short amplicons before sequencing may be desired,” *id.* at 4, is an express invitation to a person of ordinary skill in the art to find and incorporate such a removal step in lieu of avoiding the production of short amplicons.

As discussed above, Lao teaches a method of removing short amplicons from a reaction. Specifically, Lao discloses “an event or state in which a portion of a nucleic acid strand is hybridized to another portion of itself,” which Lao refers to as “self-hybridizing.” Ex. 1014 ¶ 69. This “self-hybridized structure [may only be] formed for the shorter insert sections,” allowing for “selective amplification of longer insert sections over shorter

insert sections.” *Id.* ¶ 116. This happens because the shorter insert sections “take themselves out of a reaction” through self-hybridization. *Id.*

Because Gardner expressly teaches combining its method with “a step to remove short amplicons before sequencing,” and because Lao teaches such a step, we find that a person of ordinary skill in the art would have had a reason to combine the teachings of Lao with those of Gardner. The parties raise several other arguments, including whether the combination of Lao and Gardner would in fact achieve both Gardner’s benefit of avoiding overlapping amplicons and Lao’s benefit of saving time and cost due to avoiding having to run separate reactions, but we need not resolve those disputes. The presence of an express reason to combine is sufficient.

(2) Reasonable Expectation of Success

Petitioner argues that a person of ordinary skill in the art would have had a reasonable expectation of success in combining the teachings of Gardner and Lao “because Lao’s method eliminates short amplicons.” Pet. 56. In addition, Petitioner argues that a person of ordinary skill in the art would have had a reasonable expectation of success because the methods of Gardner and Lao are similar, as well as because the two references have the same purpose, use the same technique to achieve amplification, and use the same reaction components. *Id.*; see Reply 12–20. Finally, Petitioner argues that the necessary modification to Gardner to take account of Lao’s teachings is simple. Pet. 56; see Reply 12–20. Patent Owner disagrees with all of Petitioner’s arguments regarding a reasonable expectation of success. PO Resp. 37–46; Sur-Reply 14–22. We consider each argument in turn.

The first argument is whether “Lao’s method eliminates short amplicons,” as Petitioner argues. Pet. 56. As discussed above, Lao teaches

a method by which short amplicons “take themselves out of a reaction” through self-hybridization, allowing for “selective amplification of longer insert sections over shorter insert sections.” Ex. 1014 ¶ 116. Patent Owner argues that Petitioner relies on an argument that “Lao’s method eliminates short amplicons” but that Petitioner fails to show that Lao teaches the complete elimination of all short amplicons, a necessary predicate to support Petitioner’s argument. PO Resp. 37–47. Petitioner replies that even incomplete elimination would have been sufficient to permit a person of ordinary skill in the art to reasonably expect success. Reply 12–14.

We note that the challenged claims themselves do not require complete elimination of short amplicons, but rather only the creation of a “secondary structure” that “is stable during a primer annealing step of the multiplex polymerase chain reaction.” Ex. 1001, 135:7–12. Because “[t]he reasonable-expectation-of-success analysis must be tied to the scope of the claimed invention,” *Teva Pharms. USA, Inc. v. Corcept Therapeutics, Inc.*, 18 F.4th 1377, 1381 (Fed. Cir. 2021), Petitioner need not show that a person of ordinary skill in the art reasonably would have expected to succeed in completely eliminating all short amplicons using the combination of Lao and Gardner.

Nevertheless, Patent Owner argues (1) that Petitioner chooses to rely on Lao’s alleged but unproven complete elimination of short amplicons as its reason to combine the teachings of Lao with those of Gardner and (2) that Petitioner therefore should be required to prove that a person of ordinary skill in the art reasonably would have expected that result. PO Resp. 37–47. We disagree. As discussed above, Petitioner’s reason-to-combine argument relies on Gardner’s suggestion to add to its method a step to remove short

amplicons and Lao's teaching that its method "removes short amplicons." Pet. 55–56. Petitioner's argument that Lao "removes short amplicons" falls short of an argument that Lao eliminates all short amplicons; removal of some but not all short amplicons is still the removal of short amplicons.

The conclusion that incomplete removal of short amplicons is sufficient is bolstered by the language of the '359 patent itself. The method of the '359 patent "results in less efficient amplification" of the amplicons that are "subject to stable secondary structure," not in complete elimination of those amplicons from the reaction. Ex. 1001, 7:13–17. Thus, Petitioner does not argue and need not prove that a person of ordinary skill in the art would have expected the combination of Gardner and Lao to eliminate short amplicons entirely from the reaction. The '359 patent describes and claims only the formation of a stable secondary structure that causes the amplicons subject to that structure to be less efficiently amplified. This effect, resulting in removal of some, but not necessarily all, short amplicons from Gardner's reaction, is what Petitioner argues a person of ordinary skill in the art reasonably would have expected from combining the teachings of Lao with those of Gardner. All that remains is for us to determine whether the preponderance of the evidence supports that argument.

Petitioner directs us to evidence that supports its argument that a person of ordinary skill in the art reasonably would have expected the combination of Lao and Gardner to result in the removal of some of the short amplicons from Gardner's reaction. First, as Petitioner notes, Lao teaches that, under its method, shorter amplicons will "self-hybridize faster than" longer amplicons and that the shorter amplicons "thus take themselves out of a reaction." Pet. 3, 14, 15, 65; Ex. 1014 ¶ 116. This teaching led Dr.

Bao, Petitioner's declarant, to testify that "Lao's method eliminates the short amplicons produced if Gardner's primers are all combined in the same reaction." Ex. 1002 ¶ 179; *see* Pet. 56 (citing Dr. Bao's testimony).

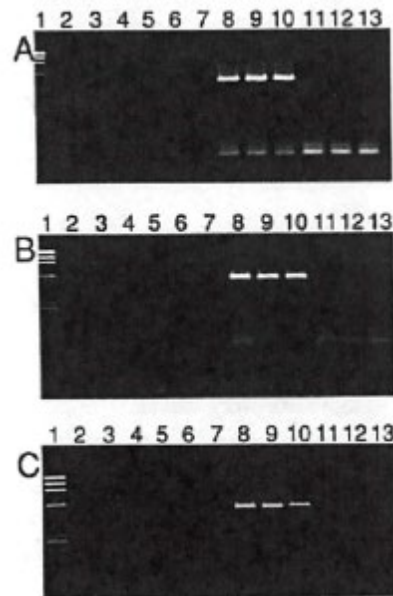
Patent Owner argues that Lao places a limit on the length of the short amplicons that can be removed by its method and that Gardner's short amplicons exceed that limit, suggesting that a person of ordinary skill in the art would not have expected the combination of Lao and Gardner to result in the removal of those short amplicons. PO Resp. 39–43. It is true that Lao teaches the selective amplification of amplicons exceeding "100 nucleotides in length" over shorter amplicons. Ex. 1014 ¶ 13. In addition, Gardner includes an example in which many overlap amplicons are longer than 100 nucleotides. Ex. 1019, 6 (listing 36 amplicons, with 30 of the 36 exceeding a "Product size" of 100). But not all of Gardner's overlap amplicons exceed 100 nucleotides: six of the amplicons listed in Gardner's example are shorter than 100 nucleotides and therefore would have been removed even under the most restrictive reading of Lao. As discussed above, Petitioner need only prove that the combination of Lao and Gardner would have been expected to remove some short amplicons. Accordingly, the fact that, even under the most restrictive reading of Lao, some of Gardner's amplicons would have been removed supports Petitioner's argument that a person of ordinary skill in the art reasonably would have expected the combination of Gardner and Lao to succeed in removing some amplicons.

Next, both parties cite the Brownie reference⁶ as supporting their respective positions on the issue of reasonable expectation of success.

⁶ Jannine Brownie et al., *The elimination of primer-dimer accumulation in PCR*, 25(16) NUCLEIC ACIDS RES. 3235, 3235–41 (1997) (Ex. 1013).

Petitioner argues that Brownie shows that adding a 5' tail to primers caused a reduction in primer dimers when compared to the use of unmodified primers. Reply 15–18. Patent Owner argues that mere addition of a 5' tail was insufficient in Brownie to reduce primer dimer formation and that instead Brownie teaches that both the addition of the 5' tail and “amplifying with a primer specific to the 5' tails” were necessary to reduce primer dimers. PO Resp. 43–44. Because Petitioner’s proposed combination of Gardner and Lao includes only the use of Lao’s 5' tails and not the additional use of a primer specific to those tails, Patent Owner argues that Brownie demonstrates that a person of ordinary skill in the art would not have expected Petitioner’s combination to be successful in removing small amplicons. *Id.*

Figure 5 of Brownie is reproduced below.



Ex. 1013, 6. Figure 5 of Brownie depicts the results of several polymerase chain reactions. *Id.* Figure 5A shows the results when conventional primers were used, Figure 5B shows the results when tailed primers were used, and

Figure 5C shows the results when both tailed primers and corresponding specific primers were used. *Id.* In Figure 5A, there is fluorescence corresponding to both the specific amplicon and to primer dimers. *Id.* In Figure 5C, there is fluorescence corresponding only to the specific amplicon. *Id.* In Figure 5B, there is clear fluorescence corresponding to the specific amplicon, and there appears to be some fluorescence corresponding to primer dimers, but there is less of the latter in Figure 5B than in Figure 5A. *Id.* Brownie describes the results depicted in Figure 5B as being “[s]imilar” to those depicted in Figure 5A. *Id.*

We find Brownie to be of limited use in supporting either party’s argument. To support Patent Owner’s position that the mere addition of 5’ tails was ineffective, Figure 5B of Brownie would need to depict results nearly identical to those in Figure 5A. It does not. Instead, the level of primer dimer fluorescence in Figure 5B is considerably less than the level of the same fluorescence in Figure 5A, suggesting that the mere addition of 5’ tails to conventional primers was sufficient to cause at least some reduction in the production of primer dimers. *Id.*; Ex. 1026, 102:16–104:13, 106:14–107:11 (Michael L. Metzker, Ph.D., Patent Owner’s declarant, testifying that Figure 5B of Brownie shows lower primer dimer production than Figure 5A). On the other hand, Dr. Metzker also testified that “the reproduction of the gels” in figures like Brownie’s Figure 5 is “not always that good, and so it’s sometimes difficult to directly look at the intensity and make comparisons,” which suggests that Figure 5 of Brownie is of little use in determining the effect of merely adding a 5’ tail to conventional primers, as Lao teaches. Ex. 1026, 102:16–25. Patent Owner cites Brownie’s qualitative description of the results depicted in Figures 5A and 5B as

“[s]imilar,” but that description does not support a finding that the results in Figures 5A and 5B were identical. *Id.* Accordingly, we give little weight to either Brownie’s Figure 5 or Brownie’s qualitative description of Figure 5 in reaching our decision.

The final piece of evidence the parties raise in support of their positions on reasonable expectation of success is Schenk.⁷ Patent Owner argues that Schenk, a 2017 article, teaches that merely adding 5’ noncomplementary sequences, as taught by Lao, to primers similar to those in Gardner was less effective at “eliminating the short amplicons” than other methods. Sur-Reply 18–19. Patent Owner’s argument responds to an argument raised by Petitioner in its Reply that the similar argument Patent Owner raised in the Preliminary Response, but not in the Response during trial, was incorrect. Reply 14–15. Because Patent Owner did not raise an argument based on Schenk in its Response, any such argument is “deemed waived.” Paper 8, 8. Petitioner’s argument in the Reply and Patent Owner’s argument in the Sur-Reply therefore do not respond to a validly raised argument, and they are therefore untimely. 37 C.F.R. § 42.23(b). Accordingly, we do not consider the parties’ arguments with respect to Schenk in reaching our decision.

Weighing all the evidence of record, we find that a person of ordinary skill in the art reasonably would have expected to succeed in forming stable secondary structures that remove at least some short amplicons from Gardner’s reaction, as well as in achieving the remainder of the subject

⁷ Desiree Schenk et al., *Amplification of overlapping DNA amplicons in a single-tube multiplex PCR for targeted next-generation sequencing of BRCA1 and BRCA2*, 12 PLoS ONE 1–16 (2017) (Ex. 2001).

matter of the challenged claims, using a combination of the teachings of Lao and Gardner.

4. *Conclusion*

As discussed above, Petitioner has shown by a preponderance of the evidence that the combination of Gardner and Lao teaches or suggests the subject matter of claims 1–6, as well as that a person of ordinary skill in the art would have had a reason to combine the teachings of these references with a reasonable expectation of success. Accordingly, Petitioner has proven that claims 1–6 are unpatentable because they would have been obvious over the combination of Gardner and Lao.

C. Asserted Anticipation by Xie or Lao

Petitioner argues that Xie anticipates claims 1–6. Pet. 28–41. In addition, Petitioner argues that Lao anticipates claims 1–6. *Id.* at 41–54. Because Petitioner has shown by a preponderance of the evidence that all of the challenged claims are unpatentable on the obviousness ground, we need not reach Petitioner’s anticipation arguments.

D. Patent Owner’s Motion to Exclude

During trial, Patent Owner “move[d] to exclude Exhibits 1023, 1025, and 1027, as well as any portions of the record relying on these exhibits.” Paper 26, 1. Patent Owner did not specify all of these additional “portions of the record” but did note that they included “at least Paper 20, pp. 12, 13–15, [and] 17–18.” *Id.*

In reaching our conclusions on Petitioner’s asserted grounds of unpatentability, we do not rely on any portion of Exhibits 1023, 1025, or 1027. Accordingly, we dismiss Patent Owner’s Motion to Exclude those exhibits as moot.

With respect to Patent Owner's Motion to Exclude all portions of the record relying on Exhibits 1023, 1025, and 1027, Patent Owner identifies six pages of Petitioner's Reply that it believes we should exclude. None of the identified pages cites any portion of Exhibits 1023, 1025, or 1027. *See* Reply 12–15, 17–18. Thus, even if we were persuaded that Exhibits 1023, 1025, and 1027 should be excluded, we would not be persuaded that Patent Owner had identified any reason for us to exclude the identified pages of Petitioner's Reply. Accordingly, we deny Patent Owner's Motion to Exclude as it applies to Petitioner's Reply. To the extent there are other portions of the record that do cite Exhibits 1023, 1025, or 1027, Patent Owner's failure to identify those portions is fatal to its motion. 37 C.F.R. § 42.22(a)(1) (a motion "must include . . . [a] statement of the precise relief requested"). Accordingly, we deny the motion as it applies to any unidentified portion of the record.

CONCLUSION⁸

For the reasons discussed above, Petitioner has shown by a preponderance of the evidence that claims 1–6 of the '359 patent are unpatentable.

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–6	102	Xie ⁹		
1–6	102	Lao ⁹		
1–6	103	Gardner, Lao	1–6	
Overall Outcome			1–6	

ORDER

It is hereby

ORDERED that, based on the preponderance of the evidence, claims 1–6 of the '359 patent have been shown to be unpatentable;

⁸ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this Decision, we draw Patent Owner's attention to the April 2019 Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. §§ 42.8(a)(3), (b)(2).

⁹ This Final Written Decision does not reach these grounds because Petitioner has proven all challenged claims are unpatentable on its obviousness ground.

FURTHER ORDERED that Patent Owner's Motion to Exclude is dismissed as moot as to Exhibits 1023, 1025, and 1027 and denied as to all other portions of the record; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of this Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2021-00401
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For PETITIONER:

Matthew I. Kreeger
Jian Xiao
Parisa Jorjani
MORRISON & FOERSTER LLP
MKreeger@mofo.com
JXiao@mofo.com
PJorjani@mofo.com

For PATENT OWNER:

Matthew S. Gibson
Ryan P. Cox
Steven D. Shipe
Mark C. Nelson
BARNES & THORNBURG LLP
mgibson@btlaw.com
rcox@btlaw.com
sshipe@btlaw.com
mnelson@btlaw.com